ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION — DECEMBER / JUNE 2024

B.TECH. CSE ENGINEERING — SEMESTER 5

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- UNIT I ---

Q.1 Compare OSI Reference Model with TCP/IP Protocol Suite. Explain Transmission Media. [14 Marks]

[OR]

Q.1 (Alt) Explain CSMA/CD and CSMA/CA MAC protocols in wired and wireless LANs. [14 Marks]

--- UNIT II ---

Q.2 Explain Error Detection and Correction: CRC polynomial division and Hamming code. [14 Marks]

[OR]

Q.2 (Alt) Describe Sliding Window protocols: Stop-and-Wait, Go-Back-N, and Selective Repeat. [14 Marks]

--- UNIT III ---

Q.3 Explain IPv4 addressing, Subnetting, and CIDR. Compare Distance Vector vs Link State Routing. [14 Marks]

[OR]

Q.3 (Alt) Describe OSPF, BGP, and ARP/RARP protocols. Compare IPv4 vs IPv6 headers. [14 Marks]

--- UNIT IV ---

Q.4 Explain TCP Three-Way Handshake, Flow Control, and Congestion Control (AIMD, Slow Start). [14 Marks]

[OR]

Q.4 (Alt) Compare TCP and UDP sockets. Explain Port numbers and Multiplexing/Demultiplexing. [14 Marks]

--- UNIT V ---

Q.5 Describe DNS hierarchy and resolution process. Explain HTTP, HTTPS, and SMTP protocols. [14 Marks]

[OR]

Q.5 (Alt) Explain Network Security basics: RSA public key cryptography and SSL/TLS handshake. [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****